Maintenance Schedule

Diesel engine
12V1600Gx1S
12V1600Gx1F
Application group 3B

MS50810/02E

A Rolls-Royce solution

Motortyp	kW/Zyl.	Anwendungsgruppe
12V1600G11F	56 kW/Zyl.	3B, continuous operation, variable load, ICXN, 50 Hz
12V1600G21F	59.6 kW/Zyl.	3B, continuous operation, variable load, ICXN, 50 Hz
12V1600G31F	67.2 kW/Zyl.	3B, continuous operation, variable load, ICXN, 50 Hz
12V1600G01S	55.3 kW/Zyl.	3B, continuous operation, variable load, ICXN, 60 Hz
12V1600G11S	59.3 kW/Zyl.	3B, continuous operation, variable load, ICXN, 60 Hz
12V1600G21S	63.3 kW/Zyl.	3B, continuous operation, variable load, ICXN, 60 Hz
12V1600G31S	67.4 kW/Zyl.	3B, continuous operation, variable load, ICXN, 60 Hz
12V1600G41S	75.4 kW/Zyl.	3B, continuous operation, variable load, ICXN, 60 Hz

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to §15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

Table of contents

1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product manufactured by Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of operational availability. The maintenance intervals as well as the required maintenance task scopes are based on operational experience and are therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of demanding operational and ambient conditions.

Observance of the specified maintenance intervals is essential to maintain product safety.

The intervals at which the maintenance tasks are to be performed are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product
QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule)
QL3: Maintenance work which requires partial disassembly of the product
QL4: Maintenance work which requires complete disassembly of the product

Additional notes on maintenance and preservation:

Maintenance activities stipulated in any documentation provided by third-parties (suppliers and component manufacturers), particularly those relating to individual components and subsystems, may differ from the information specified in this Maintenance Schedule. The Maintenance Schedule provided by Rolls-Royce Solutions shall take precedence in such case.

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

Refer to the applicable Fluids and Lubricants Specifications for details of fluid and lubricant change intervals.

If the product is to remain out of service for a longer period of time, Rolls-Royce Solutions recommends carrying out a monthly test run until the engine has reached its normal operating temperature. If the engine is scheduled to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications.

The latest version of the specifications is available at http://www.mtu-solutions.com.

TIM-ID: 0000046281 - 014

1.2 Special conditions

Designation	Value
Load factor	≤ 75%
Operating hours	Unlimited hours
Time between major overhaul	12,000 h

Table 2: Special conditions

Note 01:
The engines are certified for operation with the fuels approved in the Fluids and Lubricants Specifications. For operation with a high sulfur content in the fuel, the following must be observed: • The component TBO specified in the maintenance schedule refers to engine operation with diesel fuel with a sulfur content < 15 ppm, e.g. EN 590 or ULSD. If a fuel with a sulfur content > 15 ppm is used, the times specified in the maintenance schedule for component TBO of the cylinder head may be shorter. For reduced TBOs, refer to the Fluids and Lubricants Specifications. Engines with exhaust gas recirculation and/or exhaust gas aftertreatment must not be operated with excessive sulfur content in the fuel. The limit values in the Fluids and Lubricants Specifications apply.

Table 3: Special conditions

Note 02:
If the engine is used in an UPS unit with corresponding hardware, the exchange interval for the cylinder heads is 500 h.

Table 4: Special conditions